

Yang Yang

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Education

University of Southern California

PhD in Mechanical Engineering

Advisor: [Hangbo Zhao](#)

Los Angeles, United States

Aug 2025 – Present

Sichuan University

B.Eng in Mechanics with Honors

GPA: 3.83/4.0

Chengdu, China

Sep 2021 – Jun 2025

Research Interests

My research interest is robotic sensing, specifically designing advanced tactile and multi-modal sensors. My work bridges the gap between hardware innovation and robotic application. By developing sensory systems that exceed human capabilities, I aim to empower robots to achieve true human-level dexterity in complex environments.

Experience

The Chinese University of Hong Kong

Research Assistant, advised by [Hongliang Ren](#)

Hong Kong SAR, China

Oct 2024 – May 2025

Tsinghua University

Summer Research Intern, advised by [Wenbo Ding](#)

Shenzhen, China

Jun 2024 – Aug 2024

Sichuan University

Teaching Assistant, advised by [Hong Zhang](#)

Chengdu, China

Feb 2024 – Jun 2024

Shanghai Jiao Tong University

Summer Research Intern, advised by [Daolin Ma](#)

Shanghai, China

Jun 2023 – Aug 2023

Honors and Awards

USC Viterbi School of Engineering Graduate School Fellowship 2025

Top 100 Undergraduate Students of Sichuan University 2025

Second Prize of Academic Scholarship at Sichuan University 2024

First Prize of Sichuan Province Mechanics Competition Individual Race 2023

First Prize of Sichuan Province Mechanics Competition Group Race (Leader) 2023

First Prize of Academic Scholarship at Sichuan University 2023

Outstanding Students of Sichuan University 2023

Publications

Biomimetic multimodal tactile sensing enables human-like robotic perception

S. Li[†], T. Wu[†], J. Xu[†], Y. Huang, Z. Zhang, H. Zhao, Q. Xu, Z. Wang, L. Ye, **Y. Yang**, C. Lyu, S. Mu, X. Wang, Z. Xie, C. Wu, X. Yu, and W. Ding

Nature Sensors, vol. 1, pp. 52–62, 2026, doi: [10.1038/s44460-025-00006-y](https://doi.org/10.1038/s44460-025-00006-y).

Conformable Vision-Based Tactile Sensor with Enhanced Soft Elastomer Design for Palpating Irregular Anatomical Surfaces

Y. Yang, T. Zhang, Y. Wang, W. Yue, T. Liu, and H. Ren

Procedia Computer Science, vol. 271, pp. 79–85, 2025, doi: [10.1016/j.procs.2025.10.114](https://doi.org/10.1016/j.procs.2025.10.114).

Vitire: A Bimodel Visuotactile Tire with High-Resolution Sensing Capability

S. Li[†], J. Xu[†], T. Wu, Y. Yang, Y. Chen, X. Wang, W. Ding, and X.-P. Zhang

IEEE/ASME Transactions on Mechatronics, pp. 1–11, 2025, doi: [10.1109/TMECH.2025.3566394](https://doi.org/10.1109/TMECH.2025.3566394).

Three-dimension Tip Force Perception and Axial Contact Location Identification for Flexible Endoscopes using Tissue-compliant Soft Distal Attachment Cap Sensors

T. Zhang[†], Y. Yang[†], Y. Yang, H. Gao, J. Lai, and H. Ren

in *Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 538–544, 2025, doi: [10.1109/ICRA55743.2025.11128801](https://doi.org/10.1109/ICRA55743.2025.11128801).

Machine Learning-and Finite Element-Based Temperature-and Rate-Dependent Plasticity Model: Application to the Tensile Behavior

B. Zhang, Y. Yang, H. Wu, Y. Zhang, Q. Wang, H. Zhang, Y. Liu, and Q. Wang

Journal of Materials Engineering and Performance, vol. 34, pp. 14975–14986, 2025, doi: [10.1007/s11665-024-10167-5](https://doi.org/10.1007/s11665-024-10167-5).

A deep learning approach for low-cycle fatigue life prediction under thermal–mechanical loading based on a novel neural network model

Y. Yang, B. Zhang, H. Wu, Y. Zhang, H. Zhang, Y. Liu, and Q. Wang

Engineering Fracture Mechanics, vol. 306, Art. no. 110239, 2024, doi: [10.1016/j.engfracmech.2024.110239](https://doi.org/10.1016/j.engfracmech.2024.110239).

Skills

Programming: C++, Python

Software: Finite Element Method (FEM), SolidWorks, MATLAB, Ubuntu, VS Code, Gazebo, ROS

Hardware: STM32 Microcontrollers, 3D Printer, Laser Cutter, Sputtering System, Reactive Ion Etcher (RIE), Plasma Cleaner

Languages: Mandarin (Native), English (Fluent)